

03_DUAL_OPERATIONAL_HARNESS_AND_MCP_SPEC.md

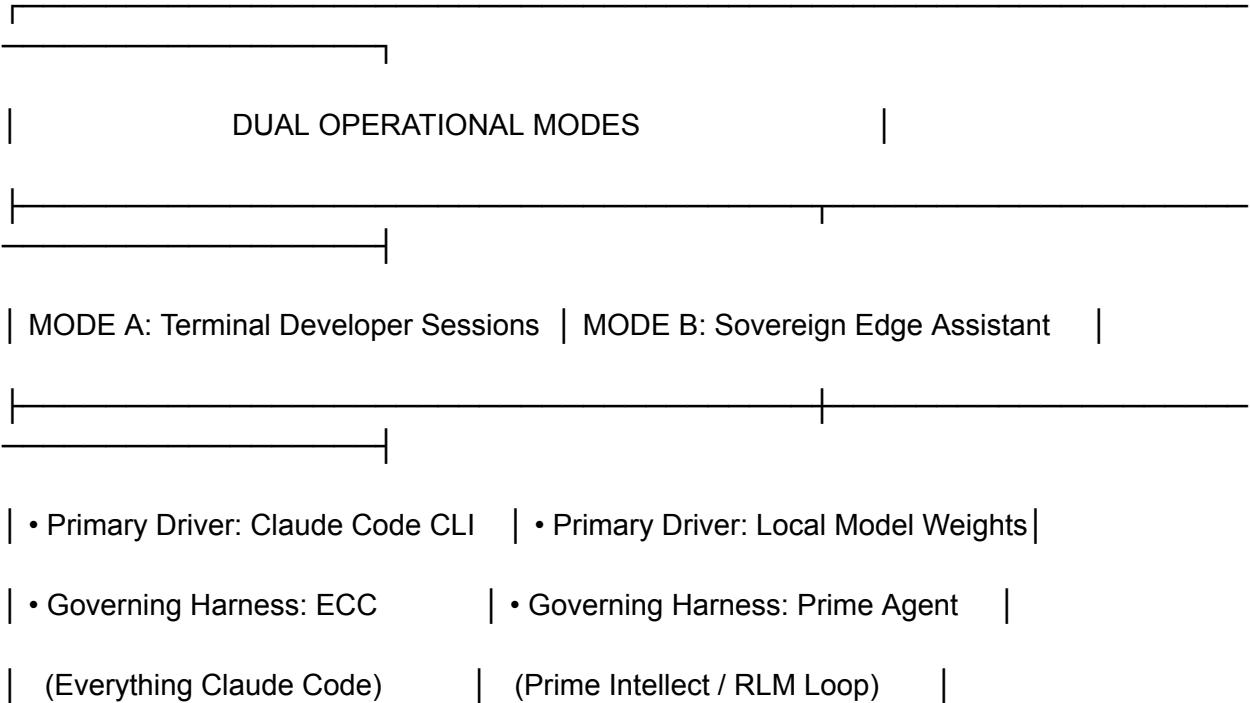
Dual Operational Modes, Prime Agent RLM Harness, MCP Infrastructure & AST Code Graphs

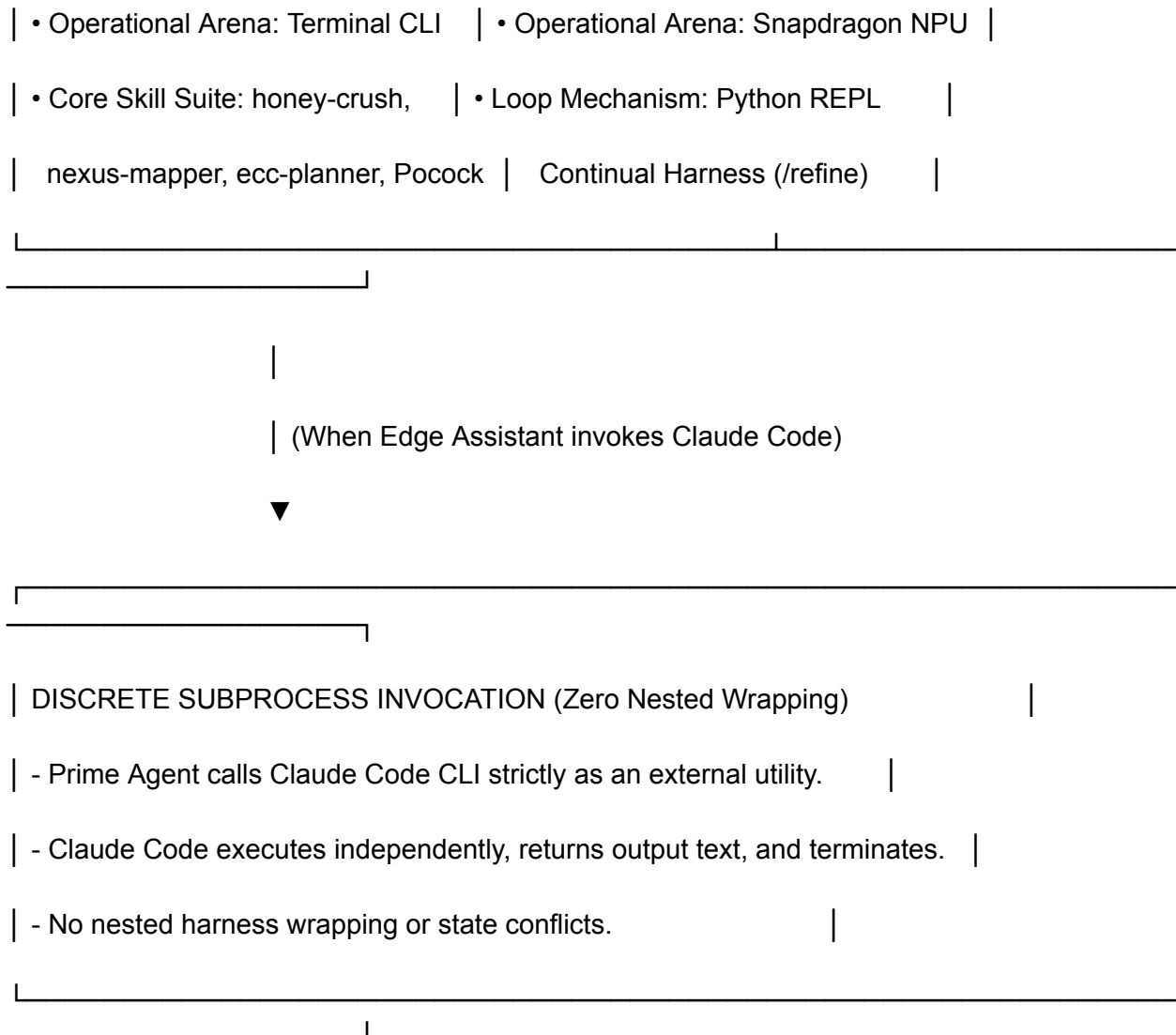
1. Document Scope & Supersession

This specification defines the execution loops, model harnesses, MCP server deployments, and code-graph analytical pipelines. It envelops, unifies, and supersedes:

- master-execution-router
- skill_manifest.json
- Data dump and skill extraction.
- Tool harness specifications in 5-Potential.34.ARCH.MATRIX-(5-S.F.'s-34-files)

2. Dual Operational Modes: Complete Separation of Concerns





Beginner-Proof Explanation:

- **Mode A (Claude Code CLI + ECC):** When you are actively sitting at a terminal typing code commands, Claude Code is your harness. ECC gives it specialized skills like `honey-crush` (reading massive docs) and `nexus-mapper` (mapping code files). Prime Agent is completely off.
- **Mode B (Edge Assistant + Prime Agent):** When you are on your phone using voice or screen vision, local open weights (Qwen 3.5) run on your Snapdragon NPU. **Prime Agent (from Prime Intellect)** runs this loop inside Python. It treats context as variables and tools as callable Python functions.
- **The Bridge:** If your on-device Prime Agent needs Claude to solve a complex coding task, it simply runs `claude --print "fix this bug"` as a command-line tool. It captures the answer and continues its day. They never fight over who owns the session.

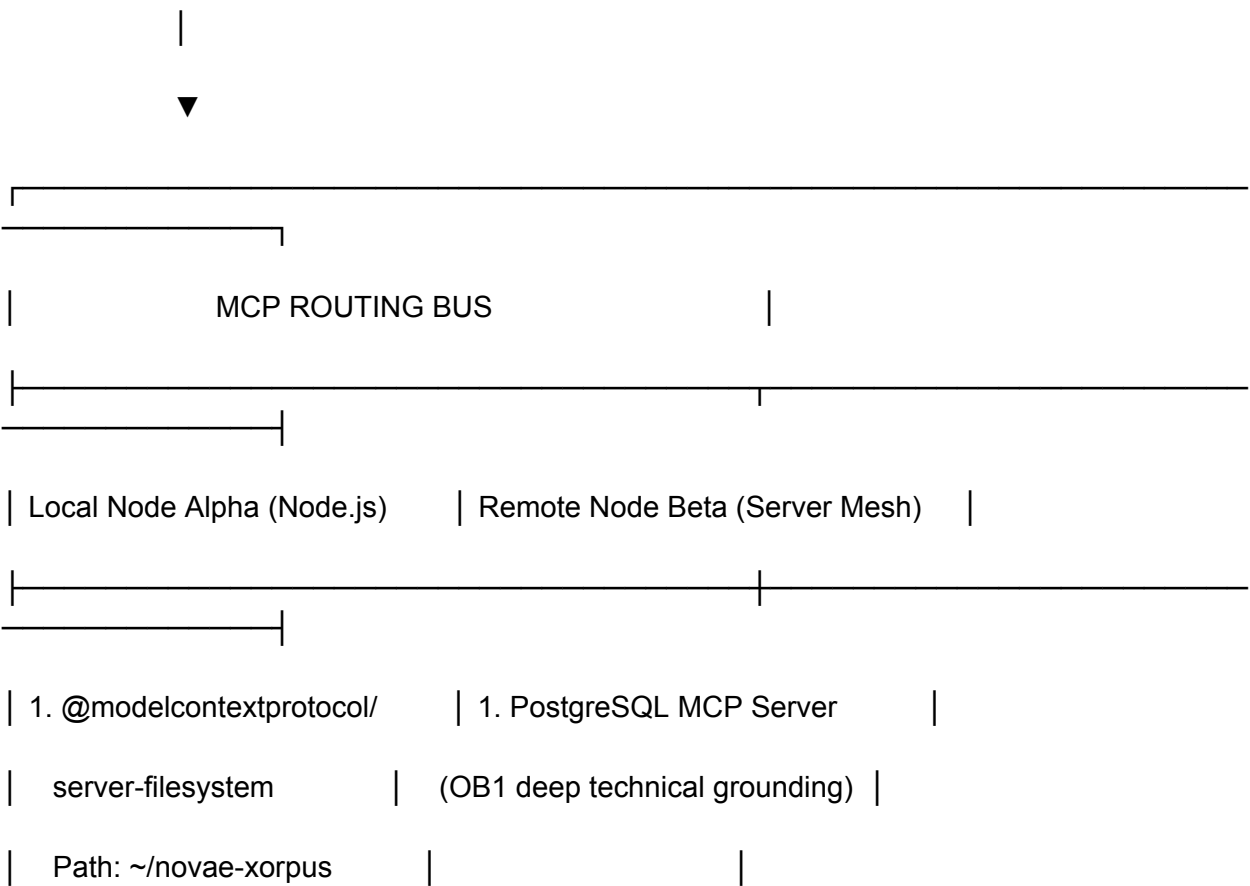
3. What Prime Agent Operates Off Of

According to Prime Intellect's architecture:

- 1. **Persistent Python / IPython Kernel:** Tools are loaded into the Python namespace as functions. The agent writes small code blocks to call them instead of massive JSON schema handoffs.
 - 2. **OpenAI-Compatible Client:** Prime Agent connects to an HTTP endpoint. It routes through **OmniRoute** on `http://localhost:20128/v1`, which handles prompt caching and token conservation before forwarding to **GenieX** on the NPU.
 - 3. **The Continual Harness:** A state layer that refines supplemental prompts and skill references via `/refine` with automated rollback snapshots.
-

4. Model Context Protocol (MCP) Server Infrastructure

[Horizons UI / Prime Agent / Claude Code]



| 2. SQLite MCP Server (Python) | 2. AST Repository Code Graph Server |

| Path: 03_recall_cache/kv_store | (PyGraphify / Graphify AST) |

- **Filesystem MCP:** Runs via Node.js LTS in the background:

```
npx -y @modelcontextprotocol/server-filesystem ~/novae-xorpus
```

Exposes the vault safely to agents without granting access to root Android system files.

5. Code Review Graph, Knowledge Graph & NotebookLM Pipeline

1. **PyGraphify / Graphify:** Ingests raw code repositories, parses Abstract Syntax Trees (ASTs), and maps which functions call which classes.
2. **OpenWiki TUI:** Writes these relationships into clean Markdown notes inside `02_wiki_md/` with bidirectional links (`[[class_a]] <-> [[class_b]]`).
3. **High-Speed Cache:** Dumps pre-tokenized chunks into `03_recall_cache/jsonl/` and SQLite tables for instant retrieval.
4. **NotebookLM (`notebooklm-py`):** Serves as the deep analytical engine for multi-document queries without flooding the prompt context window.